2025-2026 M.Sc. in Data Science and Analytics

Predicting and Explaining Traffic Collision Severity in Toronto
Using Explainable Machine Learning

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Objective

Traffic collisions remain a critical public safety concern in Toronto, causing hundreds of fatalities and thousands of injuries every year. Despite the availability of over 800,000 historical collision records from the City of Toronto Open Data Portal, most existing analyses are descriptive and fail to provide predictive insights for proactive road safety interventions. This research develops an **explainable machine learning framework** to predict traffic collision severity — classified as **Minor** (property damage only), **Major** (one or more injuries), or **Fatal** (one or more fatalities) — using **677,056** cleaned collision records (2014–2026) merged with hourly weather data from the Government of Canada's Toronto City weather station (Station ID 31688). Five classifiers (Logistic Regression, Random Forest, XGBoost, Gradient Boosting, Neural Network) are trained and compared. Five data balancing strategies address the extreme class imbalance (Fatal = 0.1%). SHAP (SHapley Additive Explanations) provides global and local model interpretations. DBSCAN spatial clustering identifies **36 fatal collision hotspot zones** across Toronto to support the city's **Vision Zero** road safety strategy.

Background

Data Sources

- City of Toronto Open Data: 809,034 collision records (2014–2026) — date, time, GPS, neighbourhood, involvement flags, injuries, fatalities
- Gov. of Canada Climate Data: hourly weather Station ID 31688 — temperature, humidity, dew point, precipitation (wind speed/visibility 100% null, excluded)
- 100% merge match on date-hour key. 131,978 invalid GPS records removed → 677,056 valid records retained with 37 engineered features

Severity Class Distribution

Severity Class	Count	%
Minor (0) — Property damage	580,492	85.7%
Major (1) — One or more injuries	95,928	14.2%
Fatal (2) — One or more fatalities	636	0.1%
Total	**677,056**	**100%**

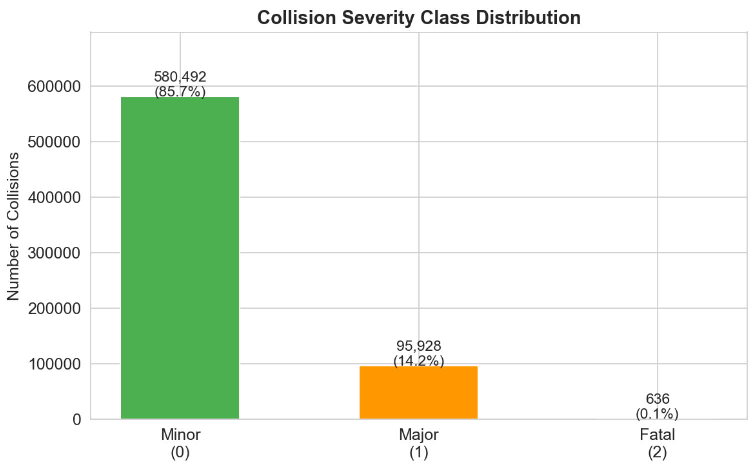

Figure 1: Severity Class Distribution

Results

Exp 1 — Model Performance (n = 101,559)

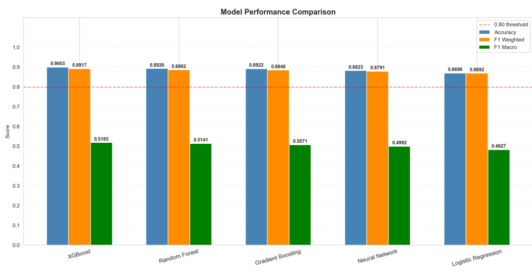

Figure 2: Model Performance Comparison

Model	Acc.	F1-W	F1-M	AUC
XGBoost ★	**0.9003**	**0.8917**	**0.5185**	**0.7839**
Rnd. Forest	0.8926	0.8862	0.5141	0.7809
Grad. Boost	0.8922	0.8848	0.5071	0.7823
Neural Net	0.8823	0.8791	0.4992	0.7624
Log. Regr.	0.8696	0.8692	0.4827	0.7452

XGBoost achieves the highest scores across all four metrics. The gap between F1 Weighted (0.892) and F1 Macro (0.519) reflects the Fatal class challenge: precision 0.02, recall 0.04 at the default threshold — motivating Experiment 3.

Exp 3 — Threshold Tuning (XGBoost)

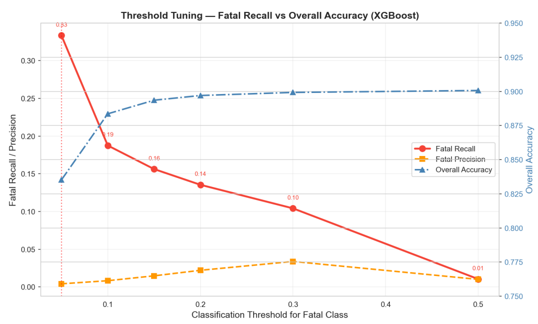

Figure 3: Lowering threshold 0.50→0.05 improves Fatal recall from 4.2% to 33.3% (8× gain)

SHAP — Top 10 Fatal Collision Risk Factors

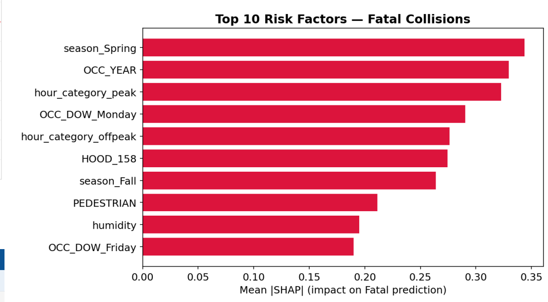

Figure 4: Top Fatal Risk Factors (SHAP) — season_Spring (0.34), OCC_YEAR (0.33), hour_category_peak (0.32) lead

season_Spring leads (0.34): pedestrians and cyclists return after winter before drivers re-adjust. **OCC_YEAR** (0.33) captures Vision Zero policy shifts (2016–2017). **PEDESTRIAN** involvement has the strongest positive directional effect (SHAP extends to +2.0). **PASSENGER** involvement is the top predictor for Major outcomes (+2.5 SHAP).

Toronto Fatal Collision Hotspot Map

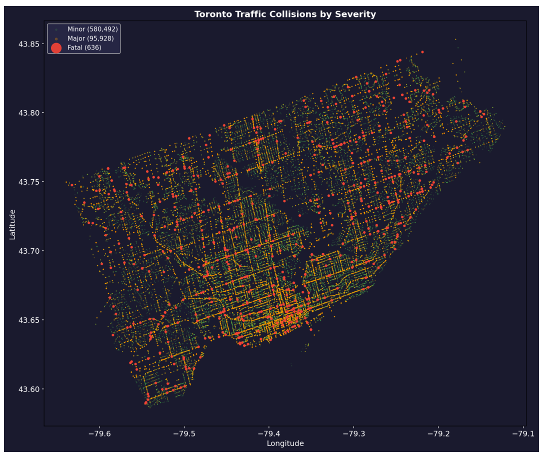

Figure 5: 36 DBSCAN hotspot zones — the road network of Toronto emerges from collision density alone

Methodology

Five-Phase Research Framework

- **Phase 1:** Load 809,034 collisions + 108,096 hourly weather records; merge on date-hour key (100% match rate)
- **Phase 2:** Remove 131,978 invalid GPS rows; fill NaN; create 3-class severity target; engineer 8 features (season, hour_category, weather flags, pedestrian/cyclist involvement, neighbourhood_freq)
- **Phase 3:** EDA — 8 visualizations covering temporal (hourly, monthly, day-of-week), spatial (neighbourhood, collision map), weather impact, and feature correlation
- **Phase 4:** Stratified 70/15/15 split (473,939 / 101,558 / 101,559); SMOTE on training set only (406,344 samples/class, 1,219,032 total); evaluate 5 classifiers on held-out test set
- **Phase 5:** SHAP TreeExplainer (2,000 test records) — global beeswarm, top-10 bar, per-prediction waterfall; DBSCAN (eps=300m, min_samples=3) on 636 fatal GPS coordinates

Balancing Strategies Compared

- No Resampling with Class Weights — penalizes minority misclassification
- **SMOTE** — synthetic minority oversampling (primary strategy selected)
- ADASYN — adaptive boundary-focused sampling
- SMOTETomek — SMOTE + Tomek Links noise cleaning
- Random Undersampling — reduce majority class to achieve balance

Data Split, Metrics & Tools

Train 70% (473,939) | Validation 15% (101,558) | Test 15% (101,559)
Primary Metric: F1 Macro | Accuracy | F1 Weighted | ROC-AUC
Tools: Python | Scikit-learn | XGBoost | imbalanced-learn | SHAP | Folium | DBSCAN

Conclusions

- **XGBoost + SMOTE is the best configuration:** Accuracy 90.0%, F1 Macro 0.519, ROC-AUC 0.784, outperforming all four competing classifiers. 5-fold CV confirms stability (Accuracy 0.892 ± 0.001) with no evidence of overfitting.
- **Threshold 0.05 recommended for deployment:** Increases Fatal recall from 4.2% to 33.3% — an 8× improvement without retraining. Missing a fatal collision is far costlier than a false alarm.
- **No single balancing strategy dominates all objectives:** SMOTE maximizes overall F1 Macro (0.519); Random Undersampling maximizes Fatal recall (72.9%). Optimal choice depends on deployment context.
- **SHAP provides actionable insights:** Top fatal predictors — season_Spring (0.34), OCC_YEAR (0.33), hour_category_peak (0.32), HOOD_158 (0.27), PEDESTRIAN involvement (0.21). Enables targeted seasonal and spatial safety campaigns.
- **36 DBSCAN hotspot zones** across Toronto — top priority: Wexford/Maryvale (18), West Humber-Clairville (17), South Parkdale (14) for safety cameras, pedestrian crossings, and speed reductions.
- **Limitations:** Wind speed & visibility unavailable (100% null); single weather station limits hyperlocal coverage; Fatal class has only 636 real records, constraining minority-class learning.

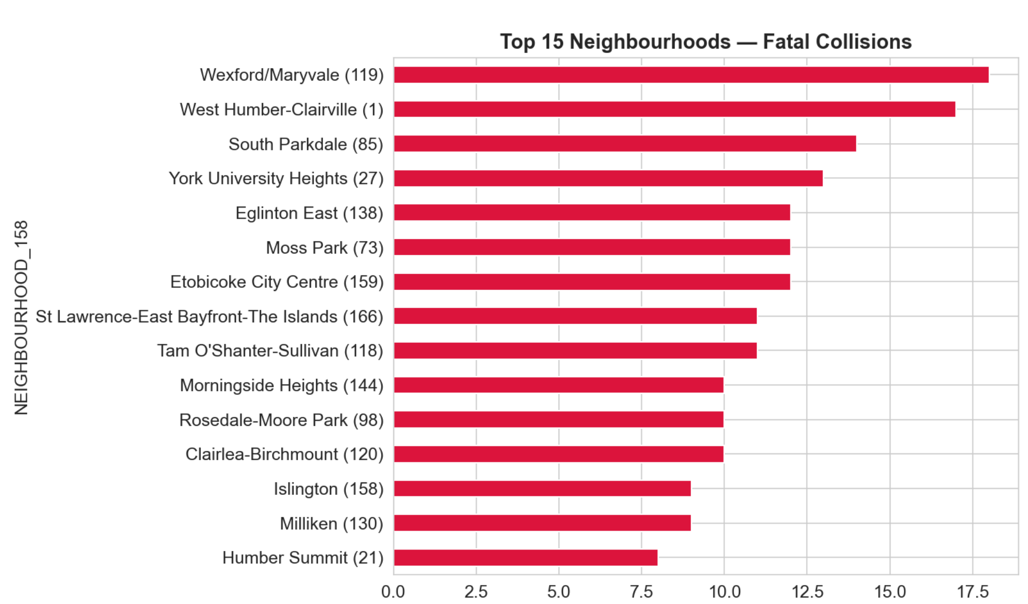

Figure 6: Top 15 Neighbourhoods by Fatal Collision Count — Wexford/Maryvale (18), West Humber-Clairville (17), South Parkdale (14)